

Time allowed; 2 Hours 30 Minutes



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READ THE FOLLOWING INSTRUCTIONS CAREFULLY

- This paper has **two** sections: **A** and **B**. Section **A** has **20** questions (**40 Marks**) and Section **B** has **12** questions. (**60 Marks**)
- Answer **ALL** questions. All the working for both sections **A** and **B** must be shown in the spaces provided.
- All working must** be done using a **blue** or **black** ball point pen or ink. Any work done in pencil other than on graphs and diagrams will **not** be marked.
- No calculators** are allowed in the examination room.
- Unnecessary **changes** in your work and handwriting that cannot be read easily may lead to **loss of marks**.
- Do not fill anything in the table indicated
"FOR EXAMINERS' USE ONLY"

FOR EXAMINERS' USE ONLY

QUESTION NUMBER	MARKS ATTAINED	INITIALS
1 - 5		
6 - 10		
11 - 15		
16 - 20		
21 - 22		
23 - 24		
25 - 26		
27 - 28		
29 - 30		
31 - 32		
TOTAL		

TERM II TOPICS AND SUBTOPICS

- TOPIC 5: FRACTIONS
- TOPIC 6: INTEGERS
- THEME 3: INTERPRETATION OF GRAPHS AND DATA
- TOPIC 7: DATA HANDLING
- THEME 4: GEOMETRY
- TOPIC 8: GEOMETRIC CONSTRUCTION
- THEME 5: MEASUREMENTS
- TOPIC 9: TIME

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APPROVED:

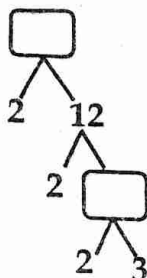
[Signature]
Consultant

Mathematics Department (PEC)

Section A. (40 Marks)

1	Workout: $36 \div 4$	2	What number has been expanded to give $(6 \times 10^3) + (9 \times 10^1) + (7 \times 10^{-2})$?
3	Given that $n(P) = 5$. Calculate the number of subsets contained in set P.	4	Find the next number in the sequence. 7, 8, 16, 43, _____
5	Given that ☆ represents 17 stars. How many stars are represented by ☆☆☆☆☆☆☆☆?	6	Write 499 in Roman numerals.
7	What number has been standardized to give 7.24×10^{-27}	8	Given that 56k is divisible by 9. Find the value of k.

9 Find the missing factors in the prime factorization below.



10 Using a pair of compasses, a ruler and a pencil only, construct the supplement of 135° at point R.



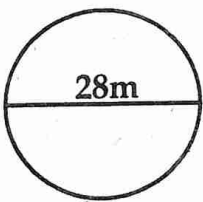
11 Workout: $123_{\text{five}} \times 4_{\text{five}}$

12 What integer is 6 steps to the right of -2?

13 Given that $a = -2$, $b = 4$ and $c = -3$, find the value of $c(a+b)$

14 Find the median of the integers below.

+3, -2, 0, -3, +1

15 WARNING: Not to be reproduced electronically!!	Otukul was facing Northwest. If he turned anticlockwise through an angle of 225°. In which direction did he finally face?	16	The letters of the word "COMPETENCE" were cut and put in a bag. Find the probability of picking letter "E" from the bag at random.
17	The exchange rate for Ksh 20 = Ugsh 600. How many Kenya shillings can be obtained from Ugsh.15, 000?	18	A car covered a distance of 60km in 40 minutes. Calculate the speed at which the car was moving.
19	Find the area of the figure below. 	20	A water tank had 720 litres of water. At breacktime, the school used $\frac{2}{3}$ of the water. How much water remained in the tank?

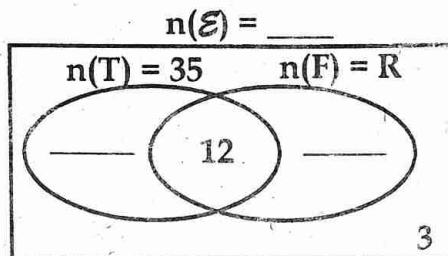
Section B. (60 Marks)

21

Kampala District athletics team was invited for a residential training at Kyambogo Sports ground. 35 athletics trained in truck events(T), R athletes trained in field events(F), 12 athletes trained in both truck and field events while 3 athletes didn't train in any of the two events.

(a) Use the above information and complete the Venn diagram below.

(02 marks)



(b) If 20 athletes didn't train in truck events at all, Find the value of R.

(02 marks)

(c) How many athletes were invited for the residential training?

(02 marks)

22

(a) If 203_b is the same as 58_{nine} , find the value of b.

(03 marks)

(b) What is the place value of 3 in 234^{five}?

(01 mark)

23

(a) Simplify: $\frac{0.48 \times 1.2}{0.08 \times 2.4}$

(03 marks)

(b) Convert $\frac{3}{8}$ to a decimal fraction.

(02 marks)

24

A school bought 8 baskets of mangoes each containing 25 mangoes. On the way, 60 mangoes got spoilt. The school decided to buy more 40 mangoes. If all the good mangoes were shared equally among 5 classes, how many good mangoes did each class get?

(05 marks)

Create a sequence of 6 terms using the rule of add 3, subtract 1 if 19 is the first term. (05 marks)

26

(a) In a basket the ratio of apples to oranges is 5 : 7. If the basket has 91 oranges, find the total number of fruits in the basket. (02 marks)

(b) If each apple is sold at sh.1500, find the total amount collected from the sale of apples. (03 marks)

27 A taxi travels from Kampala to Entebbe, a distance of 72km in 2hours.

(a) What is the speed of the taxi in m/s? (03 marks)

(b) How long would it take to cover 288km at the same speed? (02 marks)

28 A rectangular water container measures 50cm 30cm by 20cm.

(a) Find the volume of the container. (02 marks)

(b) What is the capacity of the container in litres when its $\frac{5}{6}$ full? (03 marks)

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(a) Solve: $3t - 4 = 17$

(02 marks)

(b) Sarah is 16 years older than Okello. If their total age is 60 years, how old is each of them. (03 marks)

30

When marking a test of 20 questions, a teacher awarded 3 marks for every correct answer and deducted one mark for every wrong answer.

(a) Opolot got 16 questions correct, what mark did he get? (02 marks)

(b) How many questions did a candidate who scored 28 marks fail? (03 marks)

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- 31 (a) Using a pair of compasses, a ruler and a pencil only, construct a three sided polygon ABC where $AB = BC = 6.5\text{cm}$ and angle $ABC = 60^\circ$ (04 marks)

(b) Measure line segment AC. (01 mark)

- 32 The table below shows marks scored by pupils in P.7 in End of term 2 examinations at Larnyard Junior School.

Marks scored	40	60	50	70	90	80
Number of pupils	3	5	2	2	2	1

(a) How many pupils did the examination? (02 marks)

(b) Find the mean mark of the pupils who scored between 40 and 90 marks. (03 marks)

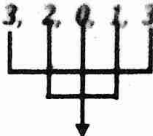
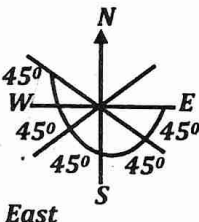
THE PRIME BEGINNING OF TERM II EXAMINATIONS 2026

P.7 MATHEMATICS MARKING GUIDE

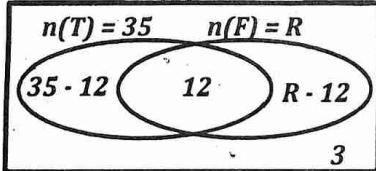
SECTION A (40 MARKS)

SECTION A (40 MARKS)

NO	SOLUTION	MAR KS	COMME NT	NO	SOLUTION	MAR KS	COMME NT						
1	$\frac{9}{36} \div \frac{1}{4} = 9$	M ₁ A ₁	Follow through	2	$(6 \times 1000) + (9 \times 10) + 7 \times \frac{1}{100}$ $6000 + 90 + \frac{7}{100}$ $6000 + 90 + 0.07$ 6000 90 $+ 0.07$ $\underline{6090.07}$	M ₁ A ₁	Follow through						
3	No. of subsets = 2^n $= 2^5$ $= 2 \times 2 \times 2 \times 2 \times 2$ $= 32 \text{ subsets}$	M ₁ A ₁	Follow through	4	$7, 8, 16, 43, 107$ $+1 + 8 + 27 + 64$	M ₁ A ₁	Follow through						
5	$17 \times 7 = 119 \text{ stars}$	M ₁ A ₁	Follow through	6	$499 =$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>400</td><td>90</td><td>9</td></tr> <tr> <td>CD</td><td>XC</td><td>IX</td></tr> </table> $= \text{CDXCIX}$	400	90	9	CD	XC	IX	B ₂	Follow through
400	90	9											
CD	XC	IX											
7	$\frac{724}{100} \times \frac{1}{10} \times \frac{1}{10}$ $= \frac{724}{1000}$ $= 0.0724$	M ₁ A ₁	Follow through	8	$5 + 6 + k = 18$ $11 + k = 18$ $11 - 11 + k = 18 - 11$ $k = 7$	M ₁ A ₁	Follow through						
9		M ₁ A ₁	Follow through	10		B ₂	Follow through						
11	$\begin{array}{r} 22 \\ 123 \text{ five} \\ \times 4 \text{ five} \\ \hline 1102 \text{ five} \end{array}$ $12 \div 5 = 2 \text{ r } 2$ $10 \div 5 = 2 \text{ r } 0$ $6 \div 5 = 1 \text{ r } 1$	M ₁ A ₁	Follow through	12	The integer is +4	M ₁ A ₁	Follow through						

13	$c(a+b)$ $3(2+4)$ 3×2 6	M_1	Follow through	14		M_1	Follow through										
15		M_1	Follow through	16	<table border="1" data-bbox="751 336 1227 376"> <tr> <td>C</td><td>O</td><td>M</td><td>P</td><td>E</td><td>T</td><td>E</td><td>N</td><td>C</td><td>E</td> </tr> </table> $n(S) = 10$ $n(D.C) = 3$ $P = \frac{n(D.C)}{n(S)}$ $= \frac{3}{10}$	C	O	M	P	E	T	E	N	C	E	B_1	Follow through
C	O	M	P	E	T	E	N	C	E								
17	$Ksh\ 20 = Ugsh\ 600$ $Ksh\ 1 = Ugsh\ \frac{600}{20}$ $1Ksh = Ugsh\ 30$ $Ksh = \frac{500}{Ugsh\ 30}$ $Ksh\ 500$	M_1	Follow through	18	$60min = 1h$ $1min = \frac{1}{60}h$ $40min = \left(\frac{2}{3} \right)h$ $S = D \div T$ $= 60km \div \frac{2}{3}h$ $60km \times \frac{3}{2} \times 1$ $= 90km/h$	M_1	Follow through										
19	$A = \pi r^2 \times h$ $A = \frac{22}{7} \times 14m \times 14m$ $A = 22 \times 2m \times 14m$ $A = 616m^2$	M_1	Follow through	20	$Water\ used = \frac{2}{3} \times 720L$ $= 2 \times 240L$ $= 480L$ $Remaining\ water = 720 - 480$ $= 240L$	B_1	Follow through										

SECTION B (60 marks)

21	(a)  (b) $R - 12 + 3 = 20$ $R - 9 = 20$ $R - 9 + 9 = 20 + 9$ $R = 29$ (c) $n(E) = 35 + 3 + R - 12$ $= 38 + 29 - 12$ $= 38 + 17$ $= 55 \text{ athletes}$	B_1 B_1 M_1 A_1 M_1 A_1 06	Follow through
22	(a) $203_b = 58_{\text{nine}}$ $(2 \times b^2) + (0 \times b^1) + (3 \times b^0) = (5 \times 9^1) + (8 \times 9^0)$ $2b^2 + 0 + 3 = (5 \times 9) + (8 \times 1)$ $2b^2 + 3 = 45 + 8$ $2b^2 + 3 = 53$ $2b^2 + 3 - 3 = 53 - 3$ <u>$2b^2 = 50$</u> $\frac{2b^2}{2} = \frac{50}{2}$ $b^2 = 25$	M_1 M_1 A_1 B_1 04	Follow through

$$b = 5$$

(b) FFF 0
2 3 4 five
fives

23 (a) $\frac{0.48 \times 1.2}{0.08 \times 2.4}$

$$\left(\frac{48}{100} \times \frac{12}{10} \right) \div \left(\frac{8}{100} \times \frac{24}{10} \right) M_1$$

$$\frac{48}{100} \times \frac{12^3}{10} \times \frac{100}{8^4} \times \frac{10}{24^1} M_1$$

$$= 3 A_1$$

(b) $\frac{3}{8} = 0.375 M_1$

$$\begin{array}{r} 0.375 \\ 8 \overline{) 30} \\ \underline{24} \\ 60 \\ \underline{56} \\ 40 \\ \underline{40} \\ 0 \end{array} M_1$$

$$\frac{3}{8} = 0.375 A_1$$

Follow through

24 No of mangoes = 8×25
= 200 mangoes
Remaining mangoes = $200 - 60$
= 140 mangoes
Total of good mangoes
 $140 + 40$
= 180
No. of good mangoes that each class got
= $180 \div 5$
= 36 books.

B₁
B₁
B₁
M₁
A₁
05

Follow through

25 (a) 19, 22, 21, 24, 23, 26

$$\begin{array}{c} \text{19} \quad \text{22} \quad \text{21} \quad \text{24} \quad \text{23} \quad \text{26} \\ \text{+3} \quad \text{-1} \quad \text{+3} \quad \text{-1} \quad \text{+3} \\ B_1 \quad B_1 \quad B_1 \quad B_1 \quad B_1 \end{array}$$

Follow through

26

A	O	TR
5	7	12
	91	

$$91 \div \frac{7}{12} M_1$$

$$91 \times \frac{12}{7}$$

$$13 \times 12$$

$$156 \text{ fruits } A_1$$

(b) $\frac{5}{12} \times 156 M_1$

$$5 \times 13$$

$$65 \text{ apples}$$

$$\text{sh} 1500 \times 65$$

$$\text{sh } 97,500 A_1$$

Follow through

27 (a) $S = \frac{D}{T}$

$$= \frac{72 \text{ km}}{2 \text{ h}} B_1$$

$$= 36 \text{ km/h}$$

$$m/s = \frac{36 \times 1000}{1 \times 3600} M_1$$

$$= 10 \text{ m/s } A_1$$

(b) 72 km take 2h M₁

$$1 \text{ km takes } \frac{2}{72} \text{ h}$$

$$288 \text{ km take } \left(\frac{2}{72} \times 288 \right) \text{ h}$$

$$= 8 \text{ hours } A_1$$

28 (a) $V = L \times W \times H M_1$

$$= 50 \text{ cm} \times 30 \text{ cm} \times 20 \text{ cm}$$

$$= 30000 \text{ cm}^3 A_1$$

(b) Capacity = $\frac{5}{6} \times 30,000 \text{ cm}^3 M_1$

$$= 5 \times 5000 \text{ cm}^3$$

$$= 25,000 \text{ cm}^3 B_1$$

$$\frac{25000}{1000}$$

$$25 \text{ litres } A_1$$

Follow through

29 (a) $3t - 4 = 17$ M_1
 $3t = 17 + 4$
 $\frac{3t}{3} = \frac{21}{3}$
 $t = 7$ A_1

(b)

O	S	T
y	y + 16	60

$y + y + 16 = 60$ M_1

$2y = 60 - 16$

$\frac{2y}{2} = \frac{44}{2}$

$y = 22$ A_1

Okello

y

22 years

$3t - 4 = 17$ M_1

$3t = 17 + 4$

$\frac{3t}{3} = \frac{21}{3}$

$t = 7$ A_1

Sarah

y + 16

22 + 16

38 years B_1

30

(a) Marks scored = $(3 \times 16) - 1(20 - 16)$
 $= 48 - (1 \times 4)$
 $= 48 - 4$
 $= 44$ marks

Let the questions he got failed be P

Marks scored = $3(20 - P) - (1 \times P)$

$28 = 60 - 3P - P$

$28 = 60 - 4P$

$28 + 4P = 60$

$28 - 28 + 4P = 60 - 28$

$4P = 32$

$\frac{4P}{4} = \frac{32}{4}$

$P = 8$

M_1

A_1

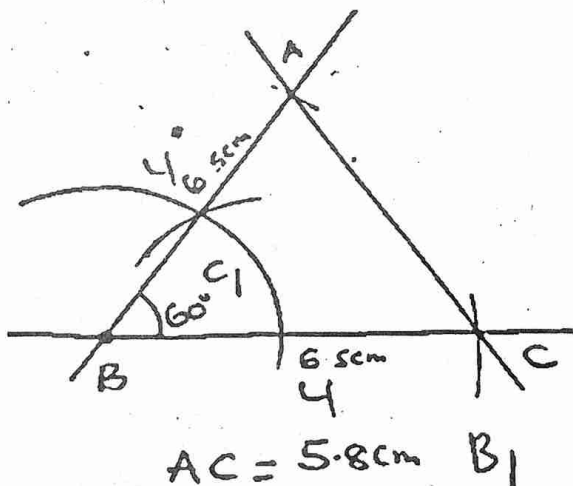
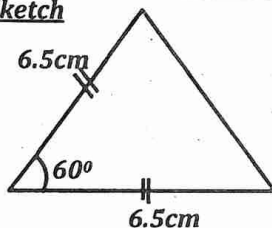
M_1

A_1

A_1

05

31 Sketch



S_1

L_1

C_1

B_1

05

32 $3 + 5 + 2 + 2 + 2 + 1$
 $8 + 7$

= 15 pupils

$(60 \times 5) + (50 \times 2) + (70 \times 2) + (80 \times 1)$

$5 + 2 + 2 + 1$

$300 + 100 + 140 + 80$

10

620

10

= 62

A_1

M_1

M_1

A_1

M_1

05